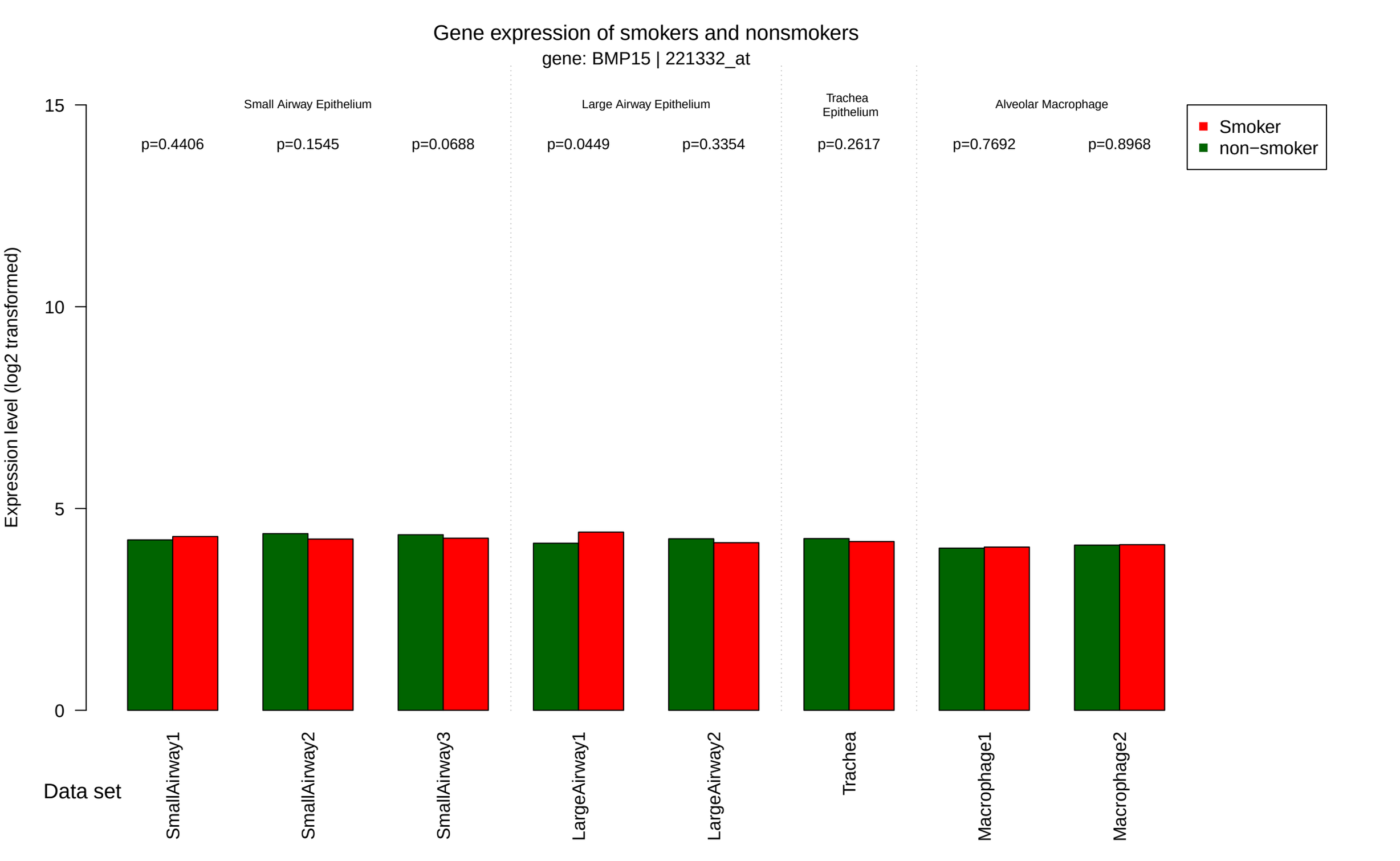
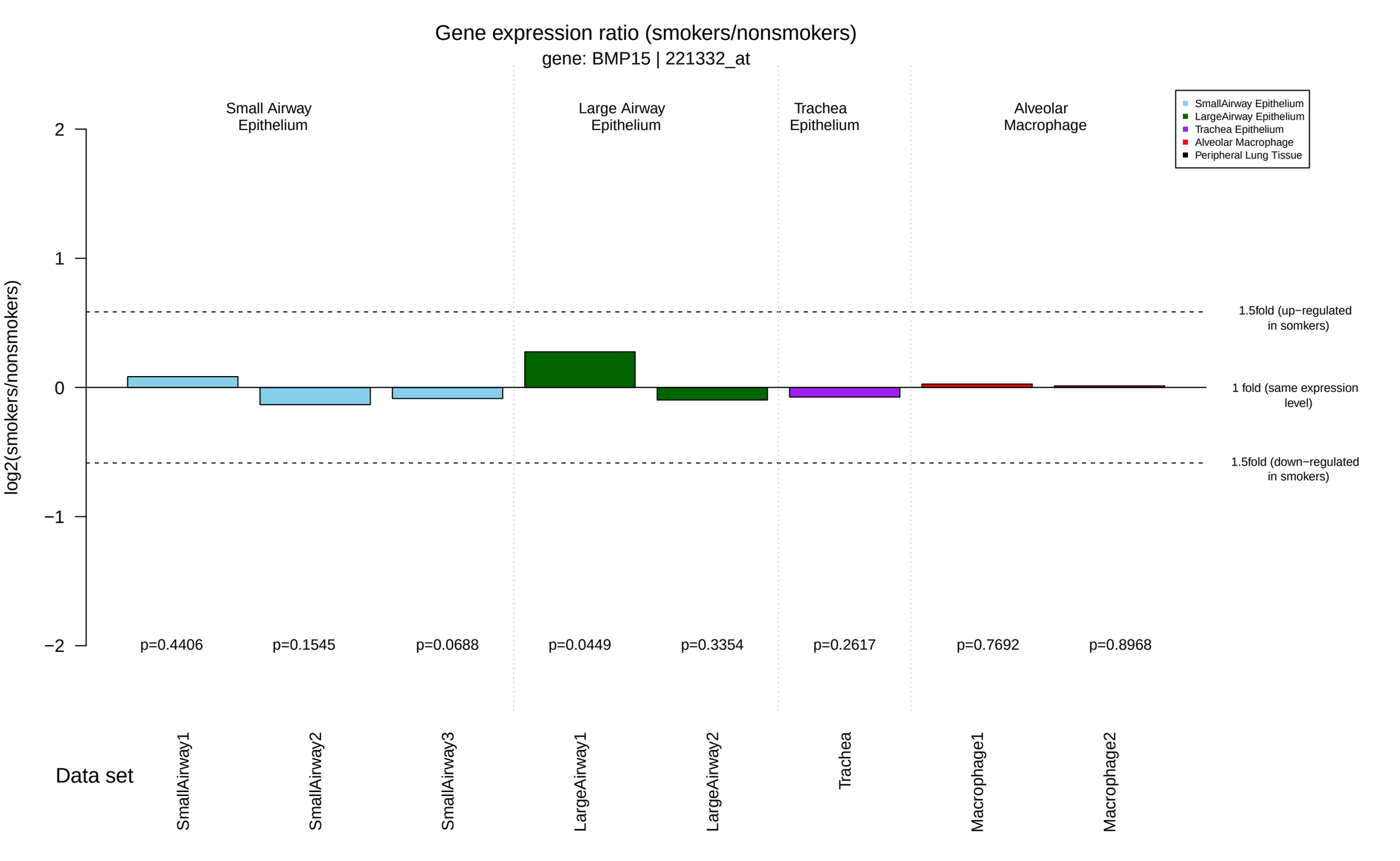
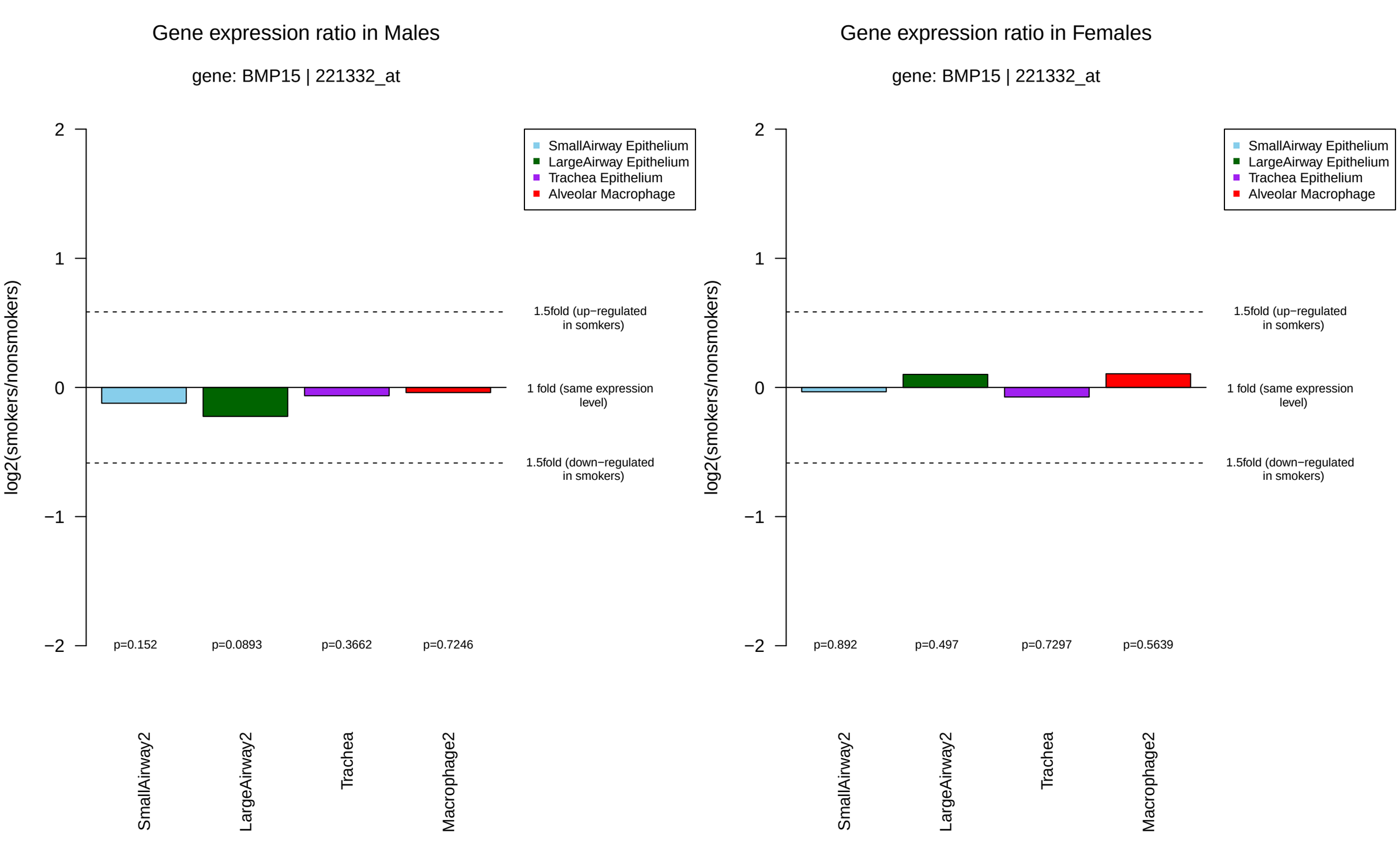
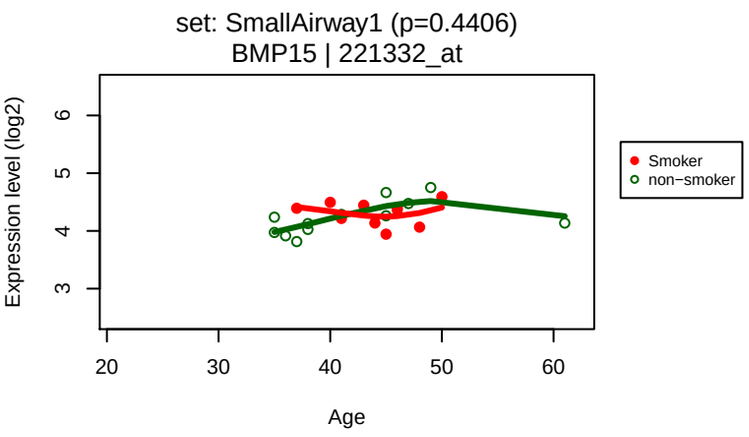
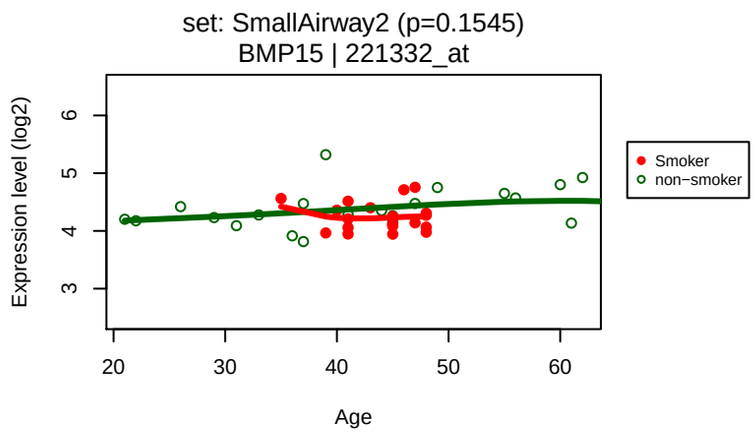
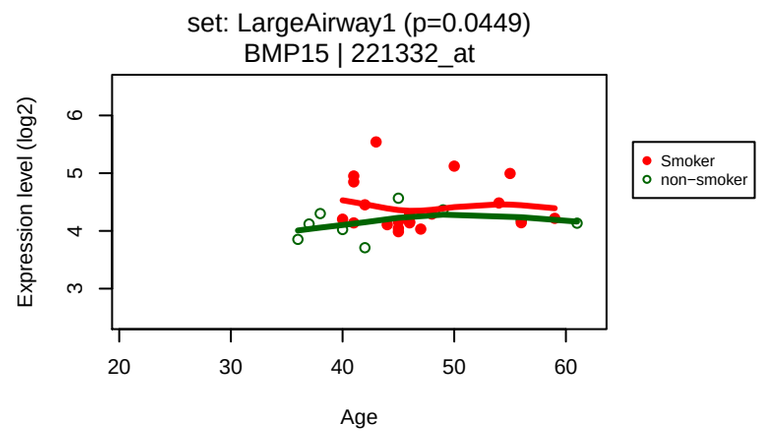

Gene expression of smokers and nonsmokers
gene: BMP15 | 221332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4406
p=0.1545
p=0.0688
p=0.0449
p=0.3354
p=0.2617
p=0.7692
p=0.8968
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

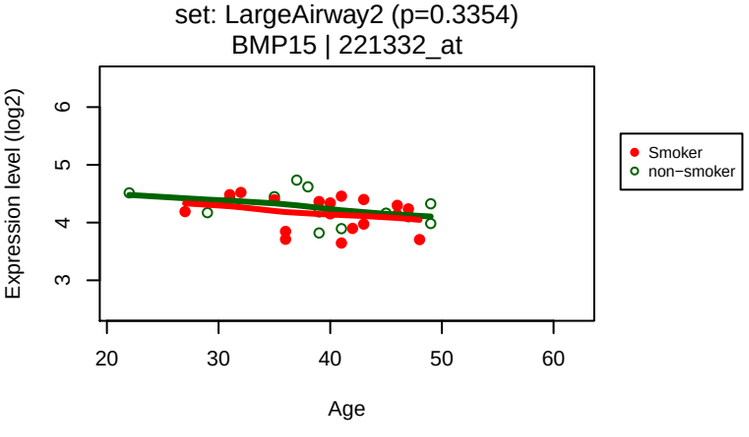

Gene expression ratio (smokers/nonsmokers)
gene: BMP15 | 221332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4406
p=0.1545
p=0.0688
p=0.0449
p=0.3354
p=0.2617
p=0.7692
p=0.8968
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

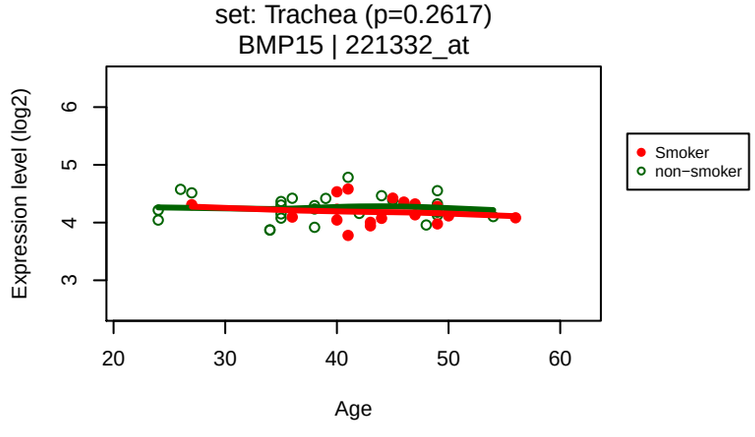

Gene expression ratio in Males
gene: BMP15 | 221332_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.152
p=0.0893
p=0.3662
p=0.7246
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP15 | 221332_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.892
p=0.497
p=0.7297
p=0.5639
SmallAirway2
LargeAirway2
Trachea
Macrophage2

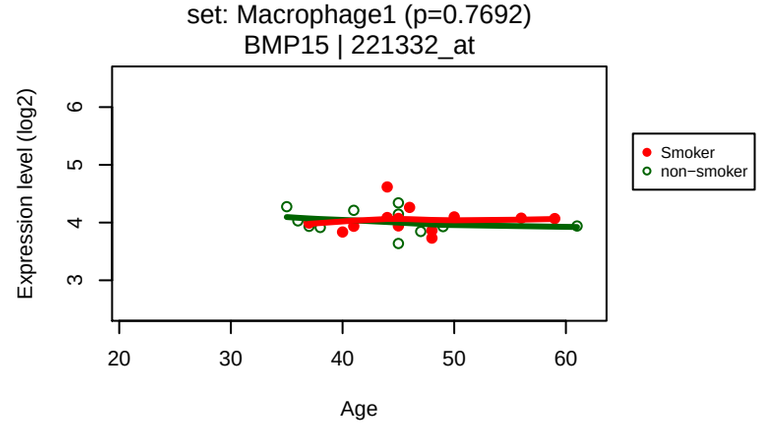

set: SmallAirway1 (p=0.4406)
BMP15 | 221332_at
Expression level (log2)
Age
Smoker
non−smoker

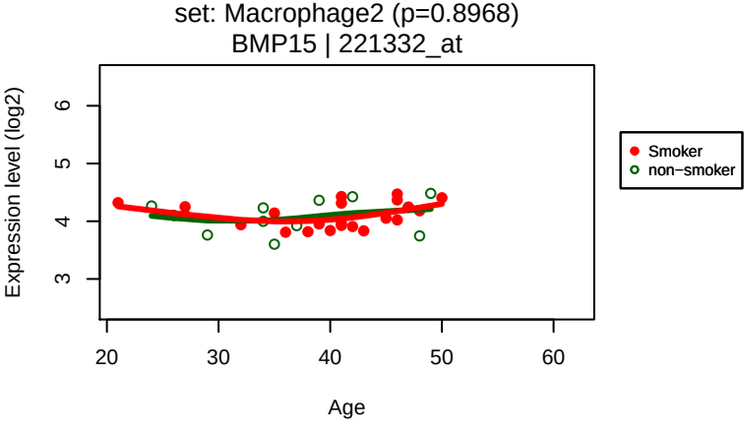

set: SmallAirway2 (p=0.1545)
BMP15 | 221332_at
Expression level (log2)
Age
Smoker
non−smoker

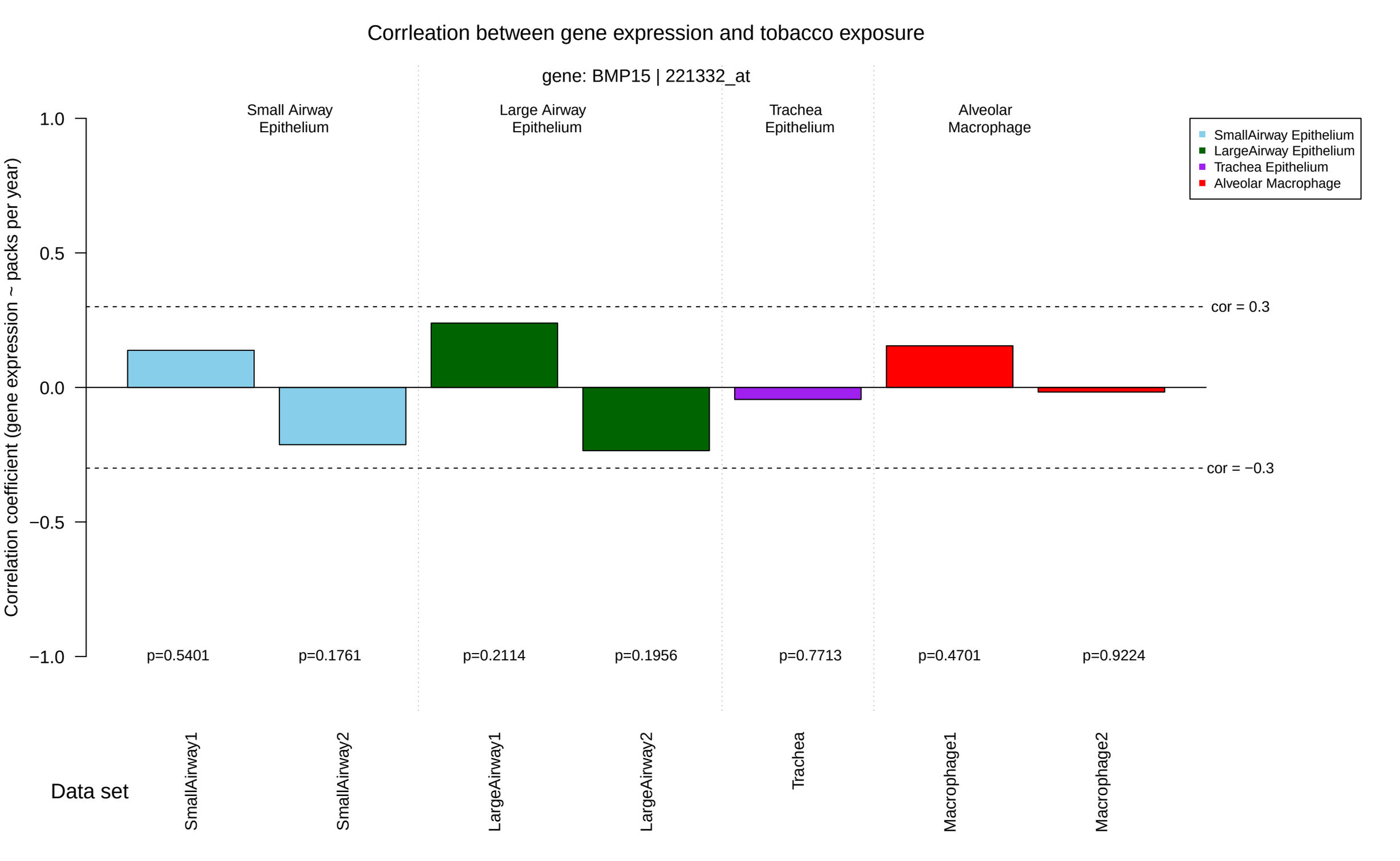

set: LargeAirway1 (p=0.0449)
BMP15 | 221332_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3354)
BMP15 | 221332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2617)
BMP15 | 221332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7692)
BMP15 | 221332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8968)
BMP15 | 221332_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP15 | 221332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5401
p=0.1761
p=0.2114
p=0.1956
p=0.7713
p=0.4701
p=0.9224
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2